ETFs AND PORTFOLIO STRATEGIES

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A bit about myself

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1. Introduction

Hedging

- Hedging is to buy or sell securities in order to guard the portfolio from adverse price movements

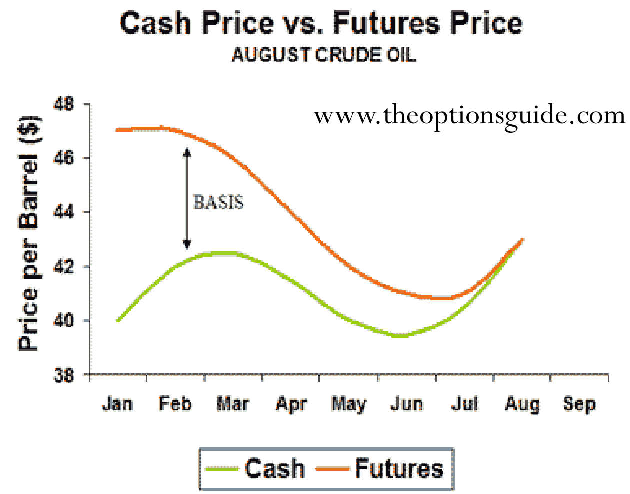
- With a portfolio of stocks (a.k.a. delta one), the hedge is normally done with futures

- With non-delta one derivatives (e.g. options), the hedge can be of different nature:
  - Delta hedge: underlying to protect vs price movements
  - Gamma (and Vega) hedge: other options to protect vs volatility changes
  - Rho hedge: bonds, swaps to protect vs rate changes

www.sharemarketschool.com

Basis Risk

- What is the basis?
  - Difference between the prices of a security and its related future



- **Contango**: Future price > underlying price (negative basis)
- **Backwardation**: Future price < underlying price (negative basis)

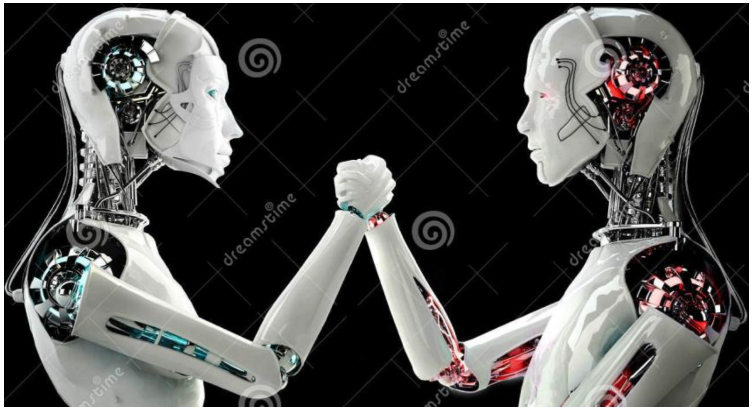
- Hedging a portfolio is not a zero-sum game
  - **Correlation risk**: The price of the future does not necessarily move as the value of the portfolio
  - **Tracking error**: Even with perfect correlation, the portfolio does not have the same weights as the index

- Examples of trading strategies that exploit the basis:
  - **Tracking**: replicate the performance of an index with a given tolerance e.g. 5% of error
  - **Rebalancing**: keep a desired level of systematic risk e.g. 80% of tracking an index
  - **Dispersion**: difference between the index volatility and single-stock volatilities

Replication

- Replicating a portfolio means reproducing its cash flows with different instruments
- Just as hedging, replicating carries a risk
  - It is not a perfect replication due to different factors: liquidity, correlation, basis

- Index funds or "trackers"
  - They try to replicate the performance of a benchmark
  - S&P 500, MSCI Mexico, DAX, etc
- **Trackers** are cheaper than traditional funds
  - Easier to track the market than beat it
  - Less transaction costs and management costs



www.dreamstime.com

- Replicating an index could be done in several ways
  - **Full replication**: buying all the constituents of the index in the right proportions
  - **Sampling**: only buying a subset of the constituents of the index based on a given tracking error tolerance e.g. 10%

- ETFs (as we will see) are examples of index funds

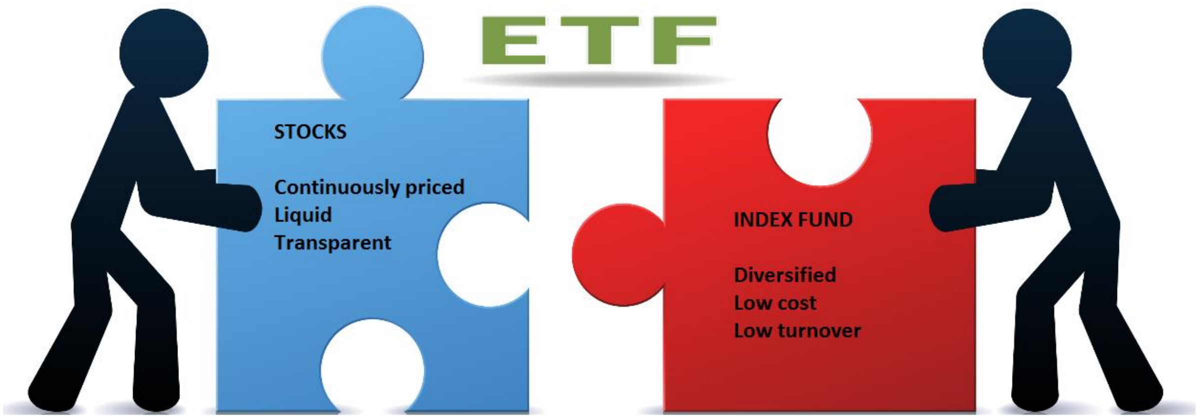
2. ETFs

Exchange Traded Funds (ETFs)

- Exchange Traded Funds are pooled **investment funds** that track a benchmark
  - Equity: equity market, sector, tailored basket
  - Bonds: treasuries, corporate, pool of sovereigns
  - Exchange Traded Commodities (ETC): oil, metals, agriculture
  - Exchange Traded Currencies (ETC): EUR, USD, MXN

- Their aim is to minimise the "tracking error" vs benchmark
  - A good ETF will replicate the performance of its benchmark

- Like stocks, ETFs have daily prices available and can be **traded on exchanges** throughout the day at current bid and ask prices



How big is the ETF market?

- 2.7tn USD invested in ETFs worldwide
  - 1.7tn USD in US Equities
  - 555bn USD in non-US Equities
  - 457bn USD in Bonds

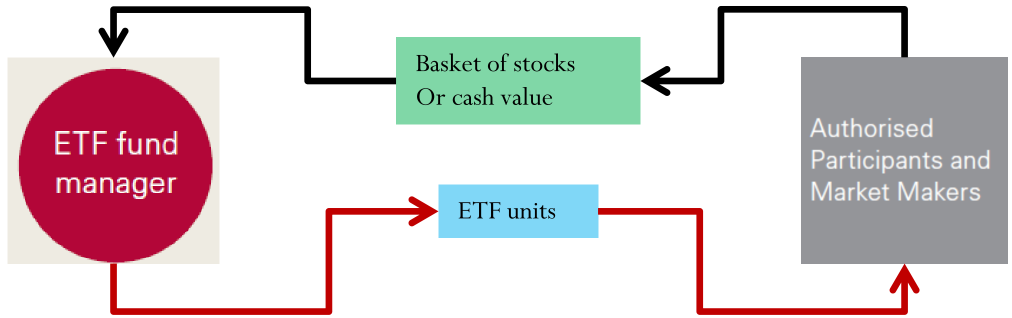
- 1,736 ETFs worldwide
  - 788 US Equities
  - 633 non-US equities
  - 289 Bonds

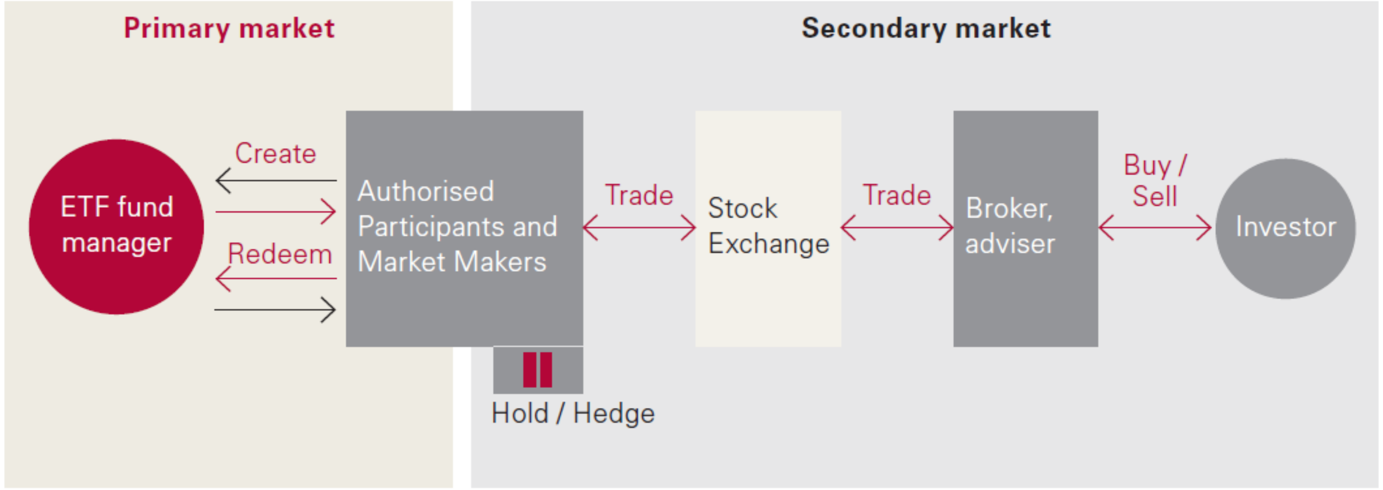
- On London Stock Exchange
  - 900+ different ETFs are traded

www.benzinga.com

Sources: www.ici.org and www.hl.co.uk

ETF creation and redemption





Sources: www.vanguard.co.uk

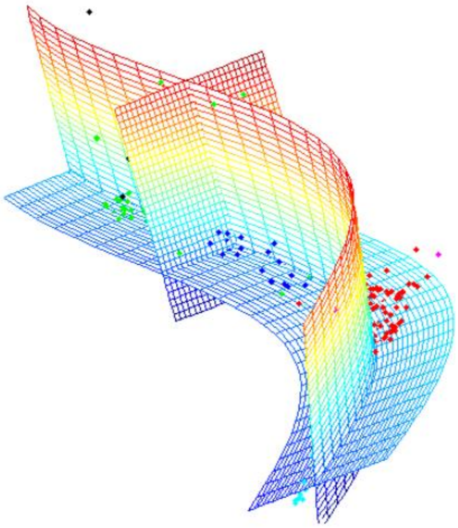
ETF indexing strategies (replication)

ETFs are derivative products:

- Their Net Asset Value (NAV) depends of the price of the constituents of the benchmark

Physically backed

- **Full replication**
  - All underlying securities are bought according to their weight in the index
  - This mean minimal tracking error vs index
  - Usually applied for smaller and liquid indices
- **Optimised sampling**
  - Only a sub-sample of the underlying securities are bought
  - Optimal number of securities based on a target tracking error e.g. 10%
  - A risk-optimised algorithm is used to compute the parameters
  - Usually applied to broader and less liquid indices



www.nlpca.org

Synthetic replication

- **Swaps**
  - The fund invests directly in derivatives e.g. swaps
  - Equity-linked swaps exchange a floating rate (LIBOR + fee) vs the return of an index

Pros: low-cost exposure

- They provide exposure to financial products and markets that would normally be too expensive or inconvenient to invest in

- Example: DAX 30 (German index) ~ 12k points
  - Normal DAX → 25 EUR per point ~ 300k EUR per contract
  - Mini DAX → 5 EUR per point ~ 60k EUR per contract
  - But one of the DAX ETFs is ~ 107 EUR per contract

db x-trackers DAX UCITS ETF + Watchlist
XDDX:TH TRADEGATE
↑ 106.73 EUR +0.36 +0.34%
As of 11:52 AM EDT 4/21/2017

- They are one of the lowest cost ways to "follow a market"
  - Traditional Asset manager: 0.5-1% fees per annum
  - Hedge fund: 1-2% fixed + 20% performance fees p.a.
  - ETF on DAX: 0.09% per annum

Fund information	
Fund name	db x-trackers DAX® UCITS ETF (DR) - Income
ISIN	LU0838782315
UCITS IV compliant	Yes
Share class currency	EUR
Investment Methodology	Direct Replication*
Portfolio Structure	Full Replication
Fund launch date	28 November 2012
Share class launch date	28 November 2012
All-in fee[1]	0.09% p.a.
Financial year end	31 December
Income treatment	Distribution
NAV per Share	EUR 111.40 (31 March 2017)
Total Fund Assets	EUR 583,414,418 (31 March 2017)
Reporting Fund	Yes

* ETF invests in all or in a representative sample of securities of the underlying index.

Sources: www.bloomberg.com and www.db.com

Pros: leverage and short

- **Leveraged ETFs**:
  - 2x to 5x the performance of the benchmark
  - Good for short term bets (a few days): your gains get amplified!
  - Bad if your bet is wrong: your losses get amplified!

SG Issuer DAX X5 Daily Long (5DEL)

Sell: **£46.73** | Buy: **£46.93** | ↔ **No change**

Market closed | Prices as at close on 21 Apr 2017 | Preferences

- **Inverse ETFs**:
  - You get the inverse return of the benchmark
    - If benchmark goes up, you gain
    - If benchmark goes down, you lose
  - In other words, you buy the ETFs to short the index
    - Helpful when there are short-selling restrictions (retail and institutional investors)

SG Issuer DAX X5 Daily Short (5DES)

Sell: **£18.12** | Buy: **£18.20** | ↓ **£0.32 (1.72%)**

Market closed | Prices as at close on 21 Apr 2017 | Preferences

Sources: www.hl.co.uk

Pros: Access to foreign markets

- You can invest on any world index via your local broker
- You can choose several currencies for the same benchmark

db X-trackers MSCI Mexico Index UCITS ETF GBP (DR) (XMEX)

Sell: **379.20p** | Buy: **380.30p** | ↑ **1.95p (0.52%)**

Market closed | Prices as at close on 21 Apr 2017 | Preferences

db X-trackers MSCI Mexico Index UCITS ETF USD (DR) (XMES)

Sell: **$4.84** | Buy: **$4.86** | ↓ **$0.0065 (0.13%)**

Market closed | Prices as at close on 21 April 2017 | Preferences

db X-trackers Nikkei 225 UCITS ETF (XDJP)

Sell: **1,359.50p** | Buy: **1,361.00p** | ↑ **9.50p (0.70%)**

Market closed | Prices as at close on 21 Apr 2017 | Preferences

iShares VII plc Dow Jones Industrial Average UCITS ETF (CID1)

Sell: **17,128.00p** | Buy: **17,153.00p** | ↔ **No change**

Market closed | Prices as at close on 21 Apr 2017 | Preferences

Sources: www.hl.co.uk

Cons: too many choices!

- Different fund providers, markets, currencies, prices per share, "replication techniques", management fees

SPDR ETFs Europe I plc S&P 500 USD (SPY5)

Sell: **$235.53** | Buy: **$235.61** | ↑ **$0.66 (0.28%)**

Market closed | Prices as at close on 21 Apr 2017 | Preferences

db x-trackers S&P 500 UCITS (XDPG)

Sell: **3,962.00p** | Buy: **3,969.00p** | ↑ **1.50p (0.04%)**

Market closed | Prices as at close on 21 Apr 2017 | Preferences

db X-trackers S&P 500 (USD) (XSPU)

Sell: **$41.24** | Buy: **$41.27** | ↑ **$0.16 (0.38%)**

Market closed | Prices as at close on 21 Apr 2017 | Preferences

Vanguard Funds plc S&P 500 UCITS ETF USD(GBP) (VUSA)

Sell: **£34.96** | Buy: **£34.97** | ↑ **£0.19 (0.55%)**

Market closed | Prices as at close on 21 Apr 2017 | Preferences

Sources: www.hl.co.uk

Cons: riskier than stocks

www.typepad.com

- **Liquidity risk**
  - Bigger tracking error
  - More transaction costs
  - Bigger spreads
  - Harder and costlier to exit a position

- **Credit risk**
  - Replication can use complex derivatives e.g. swaps
  - The issuer of the swap may not honour their contract

- **Stock-lending risk**
  - In physical replication, some funds lend their holdings to investors that short-sell
  - There is a risk that the short-sellers would not give back the stocks in time

- **Sophisticated investments**
  - They are like "Do It Yourself" investment tools
  - Require more knowledge than single stocks or traditional funds (macro news, trading strategies, transaction costs)

3. Portfolio strategies with ETFs

Example 1: Market neutral

- Definition
  - The notional value of the long leg (buys) and the short leg (buys) are the same
  - In other words, the net value buys minus sells is zero
  - The short leg can be substituted by an inverse ETF

- Assumptions
  - The long leg will outperform the short leg
  - The short leg is a hedge when prices go down

www.123rf.com

- Construction
  - Buy: 1 unit of S&P 500 @ 40.18 GBP
  - Sell: MSCI Mexico (or buy inverse ETF)
    - We need to sell 40.18 GBP
    - 1 unit @ 3.803 GBP
    - 10 units → 38.03 GBP
    - "Basis risk" or "Tracking error": (40.18-38.03)/40.18 = 5.35%

Basic ETF portfolios

- The most basic portfolio is long-only with 2 securities
- For the next examples we will use:
  - An ETF on S&P 500: XDPG.L / XDPG:LN
  - An ETF on its volatility index, the VIX: VXIS.L / VXIS:LN
- Sources: www.bloomberg.com

- ✓ XDPG
- ✓ 1Y return = 16.68%
- ✓ Volatility 1 month = 7.13%

- ✓ VXIS
- ✓ 1Y return = -74.97%
- ✓ Volatility 1 month = 45.72%

Example 2: Same notional weight

- Definition
  - The notional value of each constituent is the same

- Assumptions
  - The long leg will outperform the short leg
  - The short leg is a hedge when prices go down

- Construction
  - 1 unit of S&P 500 @ 40.18 GBP
  - A certain number of units of VIX
    - We need to buy 40.18 GBP
    - 1 unit @ 0.117 GBP
    - 343 units → 40.13 GBP
    - Basis risk" or "Tracking error": (40.18 - 40.13) / 40.18 = 0.12%

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Example 3: Same volatility weight

- Definition
  - The weights are such that the volatility of each constituent is the same
  - This is a particular case of "volatility neutral" or "vega hedge"

www.blogs.hrhero.com

- Assumptions
  - The size of the returns are related to the current volatility
  - If the sizes are the same then the portfolio is vega-neutral

- Construction
  - 1 unit of S&P 500 @ 40.18 GBP
    - Volatility weight (vega) in cash: 40.18 * 7.13% = 2.865 GBP
  - A certain number of units of VIX such that in GBP terms the volatilities are the same
    - 40.18 * (7.13% / 45.72%) = 6.266 GBP
    - 53 units → 6.201 GBP
    - "Net vega": (40.18*7.13% – 6.201*45.72) / (40.18*7.13%)= 1.04%

Example 4: Beta neutral

- Definition
  - The net (notional) beta of the portfolio is zero (or as close as possible)

- Assumptions
  - The betas determine how much the returns can jump relative to the market
  - Prices follow the Capital Asset Pricing Model (CAPM)

- Construction
  - 1 unit of S&P 500 @ 40.18 GBP
  - Compute the beta of VIX vs S&P 500
    - Normally we use linear regression (CAPM)
    - But as a proxy let us use the 1-year returns:
    - Beta = -74.97% / 16.68% = -4.5
  - Beta value of VIX: 40.18 / 4.5 = 8.93 GBP
    - 76 units → 8.892 GBP
    - "Net Beta": (40.18*1 + 8.892*(-4.5)) / 40.18 = 0.0014

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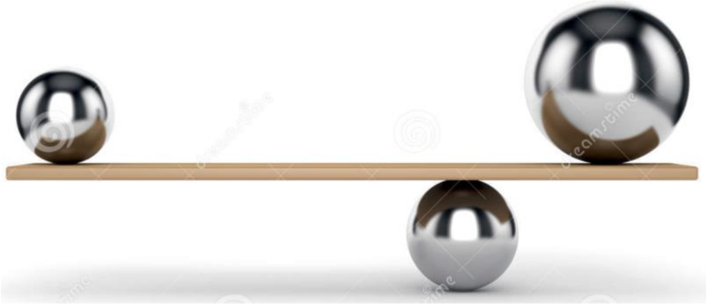
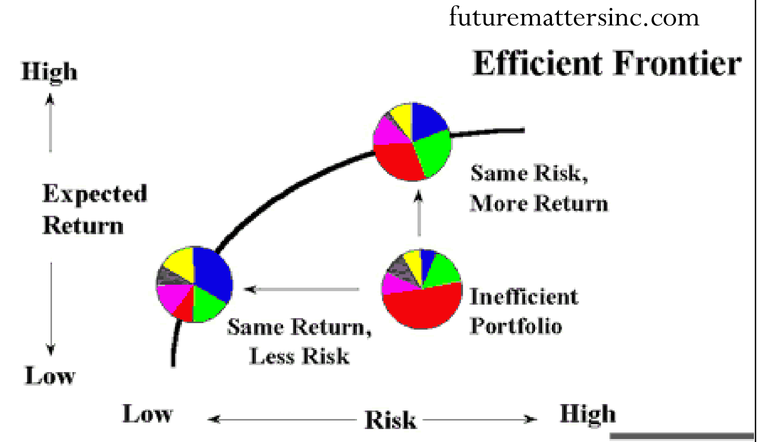


Example 5: Markowitz

- Definition: Efficient Frontier
  - Given a fixed return for the portfolio, compute the weights that minimise the variance
  - Given a fixed variance, compute the weights that maximise the return

- Assumptions
  - The variance-covariance matrix contains all risk factors



- Construction
  - 1 unit of S&P 500 @ 40.18 GBP
  - Compute the weight of VIX for a target return of 10%
    - Equation 1: $w_{S\&P} + w_{VIX} = 1$
    - Equation 2: $r_{S\&P} * w_{S\&P} + r_{VIX} * w_{VIX} = 10\%$
    - 2 equations and 2 unknowns → no need for variance-covariance matrix!
    - $w_{S\&P} = 40.18$ GBP
    - $w_{VIX} = \frac{10\% - 16.68\% * 40.18}{-74.97} = 8.806$ GBP (close to Beta neutral = 8.892)
    - 75 units → 8.775 GBP
    - "Tracking error": (40.18*1 + 8.892*(-4.5)) / 40.18 = 0.14%

4. Conclusions

Summary and conclusions

- Summary
  - We revisited the concepts of portfolio hedging and basis risk
  - We defined an ETF
  - We described several characteristics of ETFs
  - We built some examples of portfolios using ETFs

- Conclusions
  - ETFs are investment funds that can be traded in exchanges
  - They are low cost investment securities and derivative products
    - Cheaper than futures and traditional investment funds / trackers
  - They have several pros and cons
    - More liquid and more transparent prices than funds
    - More diversity and options than single stocks: leverage, inverse, asset classes, benchmarks
    - More risk: leverage, inverses, research, transaction costs

Thank you for your attention

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